

MineTrac AI — Hardware Deployment & Cloud Runbook (SIH26)

Complete 2-Node ESP32 Setup, ESP-NOW Protocol, Master Gateway & Live Dashboard Integration

■ Executive Overview:

This runbook documents the complete end-to-end hardware pipeline for the **2-Node ESP32 Hardware Setup** (Slave Node with MPU6050, ADXL345, DS3231 RTC → Master Gateway ESP32 → Render Cloud Dashboard). It includes resolved TinyML edge inference bug fixes, C header source code, FastAPI live ingestion schemas, and step-by-step flashing instructions for Rishi and the hardware team.

1. System Architecture & Data Flow

Component Overview & Network Hops:

- **Slave Node (ESP32 Dev Module):** Connects to MPU6050 (0x69), ADXL345 (0x53), and DS3231 RTC. Calculates tilt angle (pitch & roll), vibration RMS vector magnitude, applies a Stage 1 pre-filter rule (*tilt* > 3.5° or *vib_rms* > 1.8 m/s²), packs the SensorPayload struct, and transmits every 1 second via **ESP-NOW**.
- **Master Gateway Node (ESP32 Dev Module):** Listens for ESP-NOW packets from the Slave Node. Connects to an Android Phone Hotspot WiFi in STA mode, formats incoming packet data into JSON, and sends an **HTTPS POST** request to the deployed cloud endpoint (<https://mine-subsidence-ml-2rr5.onrender.com/api/ingest>).
- **Render Cloud Dashboard (FastAPI):** Stores latest node telemetry in-memory, executes Stage 1 Isolation Forest scoring, tracks a **30-second heartbeat timeout** for stale/offline node detection, and updates the Leaflet GIS satellite map automatically.

2. Summary of Resolved Edge Inference Bugs

Defect	Root Cause	Applied Fix & Verification
1. Type Truncation anomaly_detector.h	subsidence_detector_predict() returns float (-avg_depth, e.g. -4.2). Assigned to int pred_class. Checked pred_class < 0, evaluating unconditionally true for all readings.	Captured return value directly as float score. Updated check to res.anomaly_score < ANOMALY_THRESHOLD. Verified: Correct float evaluation.
2. Score Domain Mismatch train_edge_model.py	ANOMALY_THRESHOLD = -0.6127f calibrated from sklearn score_samples() (range ~[-0.8, -0.3]), but C scorer returned raw path depth (-3.0 to -8.0). Every real reading tripped as anomalous.	Calculated Isolation Forest norm factor $c(n) = 10.244771f$ for $MAX_SAMPLES = 256$. Updated C header to compute $s = -\text{powf}(2.0f, -(\text{avg_depth} / c(n)))$. Verified: FPR 1.06%, Detection 93.6%, AUC 0.9631.

3. Shared Payload Contract & Firmware Source Code

```
// esp32_firmware/shared_payload.h
#ifndef SHARED_PAYLOAD_H
#define SHARED_PAYLOAD_H
#include

typedef struct __attribute__((packed)) {
    uint32_t node_id;           // Node ID (1 to 5)
    uint32_t timestamp;         // Unix timestamp from DS3231 RTC
    float pitch;                // Pitch angle in degrees
    float roll;                 // Roll angle in degrees
    float mpu_vib_rms;           // MPU6050 vibration RMS (m/s²)
    float adxl_vib_rms;         // ADXL345 vibration RMS (m/s²)
    uint8_t stage1_anomaly_score; // 1 = Alert, 0 = Normal
    double lat;                 // Latitude fallback
    double lng;                 // Longitude fallback
} SensorPayload;
#endif // SHARED_PAYLOAD_H
```

```
// esp32_firmware/master_gateway.ino (Snippet)
void forwardToDashboard(const SensorPayload &p;) {
    HTTPClient http;
    http.begin("https://mine-subsidence-ml-2rr5.onrender.com/api/ingest");
    http.addHeader("Content-Type", "application/json");
    char jsonPayload[320];
    snprintf(jsonPayload, sizeof(jsonPayload),
        "{\"node_id\":%u,\"timestamp\":%u,\"pitch\":%.2f,\"roll\":%.2f,\""
        "\"mpu_vib_rms\":%.2f,\"adxl_vib_rms\":%.2f,\"stage1_anomaly_score\":%u,\""
        "\"lat\":%.6f,\"lng\":%.6f}\",",
        p.node_id, p.timestamp, p.pitch, p.roll, p.mpu_vib_rms, p.adxl_vib_rms,
        p.stage1_anomaly_score, p.lat, p.lng);
    int resCode = http.POST(jsonPayload);
    Serial.printf("[GATEWAY] POST to Dashboard -> Code %d\n", resCode);
    http.end();
}
```

4. Step-by-Step Hardware Flashing & Demo Runbook

Step 1: Flash Master Gateway First & Get MAC Address

1. Open `esp32_firmware/master_gateway.ino` in Arduino IDE.
2. Update the phone hotspot credentials:
`const char* WIFI_SSID = "YOUR_PHONE_HOTSPOT_SSID";`
`const char* WIFI_PASSWORD = "YOUR_HOTSPOT_PASSWORD";`
3. Flash the Master ESP32. Open Serial Monitor at **115200 baud**.
4. Copy the printed MAC address from boot logs, e.g.:
[CRITICAL] Master MAC Address: 24:0A:C4:XX:XX:XX

Step 2: Flash Slave Node

1. Open `esp32_firmware/slave_node.ino` in Arduino IDE.
2. Set hotspot credentials and paste the Master MAC address into `masterAddress[]`:
`uint8_t masterAddress[] = {0x24, 0x0A, 0xC4, 0xXX, 0xXX, 0xXX};`
3. Connect MPU6050 (AD0 to 3.3V → 0x69), ADXL345 (0x53), and DS3231 RTC.
4. Flash the Slave ESP32.

Step 3: Venue Demo Readiness & Pre-Warming Checklist

1. Enable Android Phone Hotspot (ensure 2.4GHz band is active).
2. Power on both Slave and Master ESP32 modules.
3. **1 Minute Before Demo:** Open <https://mine-subsidence-ml-2rr5.onrender.com/api/health> in browser to pre-warm the Render free-tier container and avoid cold-start delay.
4. Open the main live dashboard: <https://mine-subsidence-ml-2rr5.onrender.com/>.
5. **Live Action:** Tilt the Slave sensor (> 3.5°) or apply vibration (> 1.8 m/s²). The node marker on the GIS map turns **RED** (ALERT / ANOMALY) in real time!

5. Troubleshooting & Contingency Plan

• Issue: ESP-NOW Packets Not Arriving at Master

Cause: MAC address mismatch or WiFi channel drift.

Fix: Ensure Slave connects to phone hotspot at boot to auto-sync its WiFi channel with the Master, and verify `masterAddress[]` matches byte-for-byte.

• Issue: Render Dashboard Shows 404 / Slow Response

Cause: Free-tier container idling after 15 minutes of inactivity.

Fix: Hit `/api/health` 60 seconds before your presentation slot.

• Issue: Master HTTP POST Fails (Connection Error)

Cause: Phone hotspot lacks cellular data internet connectivity.

Fix: Verify phone has mobile data enabled, or switch to laptop local server (<http://:8050/api/ingest>).